



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO1

Case Study Topic: AgroChain – Smart Agriculture Supply Chain Management System.

Work Done Summary:

In this case study, we implemented an AVL Tree for managing agricultural products in the AgroChain Smart Agriculture Supply Chain Management System.

The AVL Tree stores product IDs in a balanced binary search tree structure, enabling fast insertion, deletion, and searching operations. We inserted multiple product records into the AVL Tree and observed balancing operations such as LL, RR, LR, and RL rotations whenever the balance factor exceeded the allowed limit.

The implementation was developed in Java to efficiently organize agricultural product information and improve retrieval performance.

Implementation Details:

- Created an AVL Tree data structure to store agricultural product IDs.
- Inserted product records into the AVL Tree in different orders.
- Calculated the balance factor for each node after every insertion.
- Performed LL (Left-Left) rotations when the tree became left-heavy.
- Performed RR (Right-Right) rotations when the tree became right-heavy.
- Applied LR (Left-Right) and RL (Right-Left) rotations when required to maintain balance.

- Implemented search functionality to quickly locate a specific product ID.
- Used in-order traversal to display product records in sorted order.
- Verified that the AVL Tree remained balanced after all insertions.

Input Screenshot :

```
class Node {
    int data;
    Node left, right;

    Node(int data) {
        this.data = data;
        left = right = null;
    }
}

public class BST {
    Node root;

    Node insert(Node root, int key) {
        if (root == null)
            return new Node(key);

        if (key < root.data)
            root.left = insert(root.left, key);
        else
            root.right = insert(root.right, key);

        return root;
    }
}
```

```
void inorder(Node root) {
    if (root != null) {
        inorder(root.left);
        System.out.print(root.data + " ");
        inorder(root.right);
    }
}

public static void main(String[] args) {
    BST tree = new BST();
    tree.root = tree.insert(tree.root, 50);
    tree.insert(tree.root, 30);
    tree.insert(tree.root, 70);
    tree.insert(tree.root, 20);
    tree.insert(tree.root, 40);

    tree.inorder(tree.root);
}
}
```

Output Screenshot :

```
AVL PRODUCT MANAGEMENT SYSTEM

Insertion Order:
1050, 1030, 1070, 1020, 1040, 1060, 1080

Rotations Performed:
No rotations required

FINAL AVL TREE

      1050
     /  \
    1030  1070
   /  \  /  \
  1020 1040 1060 1080

IN-ORDER TRAVERSAL:
1020 1030 1040 1050 1060 1070 1080

SEARCH PRODUCT ID = 1060
Product Found

Process Finished Successfully
```

Result :

The AVL Tree successfully stored and managed agricultural product records while maintaining a balanced structure. Product IDs were displayed in sorted order using in-order traversal. Search operations were completed efficiently, and balancing rotations ensured optimal tree height. The implementation demonstrated that AVL Trees provide faster search, insertion, and deletion operations compared to an unbalanced Binary Search Tree.

Time Complexity:

- Insert: $O(\log n)$
- Search: $O(\log n)$
- Delete: $O(\log n)$

Space Complexity:

- $O(n)$

GITHUB LINK <https://github.com/Himavarshini-07/AgroChain-Smart-Agriculture-Supply-Chain-Management-System>

Student Signature	Faculty Signature